BibleGoAI: AI-Driven Synthesis of Biblical Commentary Using Multi-Stage LLM Pipelines and Retrieval-Augmented Generation

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Abstract

Centuries of biblical scholarship have produced a rich corpus of verse-by-verse commentary from theologians spanning diverse traditions and eras. However, accessing and synthesizing insights from multiple commentators remains a time-consuming task, even for dedicated students of the Bible. We present **BibleGoAI**, a system that leverages large language models (LLMs), retrieval-augmented generation (RAG), and semantic similarity scoring to automatically produce concise, source-adherent summary paragraphs—called *Brief Helpful Texts* (BHTs)—from curated multi-author commentary. Each BHT distills the most striking insights from up to nine primary commentators into a single 30–80 word paragraph, with automated quality scoring and provenance tracking via footnotes. We describe the full pipeline: quote extraction, tiered synthesis, multi-strategy retrieval, iterative generation with constraint enforcement, and a composite scoring framework combining semantic accuracy, readability, and source fidelity. We evaluate the system across 7,957 New Testament and 10,492 Old Testament verses, comparing multiple generation strategies and preliminary cross-model evaluation. The resulting BHTs are served to hundreds of monthly users through BibleGo.org, a deployed web application providing interactive commentary exploration, original-language word study, and commentator browsing.

1 Introduction

The study of the Bible has generated one of the largest and most enduring bodies of scholarly commentary in human history. Prominent theologians—from John Calvin in the sixteenth century to Archibald T. Robertson in the twentieth—have produced detailed, verse-by-verse analyses that illuminate the linguistic, historical, and theological dimensions of the biblical text. For many Bible studiers, these commentaries are indispensable. Yet the sheer volume of material presents a practical challenge: consulting even a handful of commentators for a single verse requires navigating multiple lengthy texts, synthesizing sometimes divergent perspectives, and extracting the most relevant insights.

BibleGoAI grew out of a simple goal: *help people study the Bible using time-tested commentaries presented in a simple way that even new believers can use*. The project was initiated and guided by a group of Christians—mostly living in the Cleveland, Ohio area—who have been ministering to churches for years. Through weekly round table meetings, this group of volunteers and full-time ministry leaders brainstormed how artificial intelligence could make quality biblical commentary more accessible, studied the needs of modern Bible readers, and analyzed existing platforms such as BibleHub, BlueLetterBible, and StudyLight to understand how commentary is currently delivered online. The original commentary texts were sourced from StudyLight.org.

Modern advances in natural language processing (NLP) and large language models (LLMs) offer new possibilities for addressing this challenge. Systems based

on retrieval-augmented generation (RAG) have demonstrated the ability to ground LLM outputs in authoritative source documents [Lewis et al., 2020], while sentence embeddings [Cer et al., 2018, Reimers and Gurevych, 2019] enable fine-grained semantic comparison between generated and source texts. These capabilities make it feasible to build systems that not only summarize but also *validate* the fidelity of their outputs against the original material.

In this paper, we present **BibleGoAI**, a multi-stage pipeline that transforms curated biblical commentary into concise, source-adherent summary paragraphs called *Brief Helpful Texts* (BHTs). The name reflects the project's core value: bringing the richness of difficult-to-read commentary to the fingertips of Bible readers in an accessible way. The system is designed around three principles:

1. **Fidelity**: BHTs must be grounded in the actual words of trusted commentators, not in the LLM's own knowledge. At least 50% of each BHT's tokens must originate from source quotes.
2. **Conciseness**: Each BHT is constrained to 30–80 words in 1–3 sentences, making it practical for quick reference during Bible study.
3. **Traceability**: Every claim in a BHT is linked back to its source commentary via footnotes with similarity scores, enabling readers to verify and explore further.

The system has been applied to the entire New Testament (7,957 verses across 27 books) and substantial portions of the Old Testament (10,492 verses across 24 books), drawing from a curated corpus of over 20 commentators organized into three tiers of scholarly importance. The NT was publicly launched at a New Year's conference in December 2023, with the OT following at a Memorial Day conference in 2024. The resulting BHTs are deployed and actively served to hundreds of monthly users through **BibleGo.org**, a web application that provides interactive commentary exploration, original-language word study via Strong's concordance, and full commentator browsing.

Our contributions are as follows:

- A multi-stage pipeline for commentary synthesis combining LLM-based quote extraction, tiered prompt engineering, and constraint-enforced generation.
- A composite quality scoring framework that integrates semantic similarity, readability metrics, and source fidelity checks.
- A systematic comparison of four generation strategies—character-based RAG, recursive RAG, semantic RAG, and GPT-based tagging—across thousands of verses.
- A deployed, user-facing application demonstrating the practical impact of the system.

2 Related Work

Our work draws on several intersecting areas of research.

Abstractive and extractive summarization. Text summarization has been a long-standing goal of NLP, with approaches ranging from extractive methods that select key sentences [Devlin et al., 2019] to abstractive methods that generate novel text conditioned on source documents [Vaswani et al., 2017]. BibleGoAI employs a hybrid approach: the quote extraction phase is extractive (selecting verbatim passages), while the BHT generation phase is abstractive (synthesizing a coherent paragraph from selected quotes). This hybrid design allows us to leverage the fluency of LLMs while maintaining explicit grounding in source material.

Retrieval-augmented generation. RAG systems [Lewis et al., 2020] address the hallucination problem inherent in LLMs by conditioning generation on retrieved documents. Our system extends this paradigm by comparing multiple retrieval and chunking strategies—character-based, recursive, and semantic splitting—and evaluating their impact on downstream generation quality. Unlike typical RAG systems that retrieve from a single corpus, BibleGoAI retrieves from multiple independent commentaries and organizes them into a tiered structure before generation.

Semantic similarity and sentence embeddings. The Universal Sentence Encoder [Cer et al., 2018] and Sentence-BERT [Reimers and Gurevych, 2019] provide dense vector representations that enable fine-grained semantic comparison. We use these embeddings for three distinct purposes: retrieval indexing (via FAISS [Johnson et al., 2019]), quality scoring (measuring BHT-to-commentary and BHT-to-verse similarity), and footnote generation (linking BHT sentences to source quotes via cross-encoder scoring).

AI in digital humanities and biblical studies. The application of NLP to religious and historical texts is a

growing field. Prior work has explored machine translation of biblical texts [NLLB Team, 2022], named entity recognition in historical documents, and topic modeling across theological corpora. BibleGoAI contributes to this space by focusing specifically on *commentary synthesis*—a task that requires not only understanding the source texts but also respecting the scholarly authority and interpretive nuances of individual commentators.

Existing digital commentary platforms. Several widely used platforms make public-domain biblical commentary available online, including BibleHub, BlueLetterBible, StudyLight, and Logos Bible Software (commercial). These systems primarily serve as *browsers* over digitized commentary: a reader selects a verse and sees the corresponding excerpts from each commentator, one commentator at a time. They do not attempt to synthesize across commentators, enforce length budgets, or link generated text back to source quotes. BibleGoAI complements rather than replaces these tools: the source corpus itself is drawn from StudyLight, and BibleGo.org links out to full commentator text for readers who want to dive deeper. The contribution here is the *synthesis layer*—producing a short, source-grounded summary that a reader can consume in seconds rather than a multi-commentator reading session—together with the provenance mechanism that keeps each synthesized claim traceable to the underlying commentary.

3 System Architecture

BibleGoAI is composed of three major subsystems: (1) a data layer housing the biblical text and commentary corpus, (2) an AI pipeline that generates, scores, and validates BHTs, and (3) a web application that serves the results to end users. Figure 1 provides an overview of the end-to-end architecture.

3.1 Data Sources

The system operates on two primary data sources:

Biblical text. The New American Standard Bible (NASB, 1995 edition) provides the target verse text for all 66 books of the Bible. Each verse is stored as a JSON record with chapter, verse number, and text content. The NASB was selected for its reputation as a highly literal translation, which aligns well with the commentary corpus.

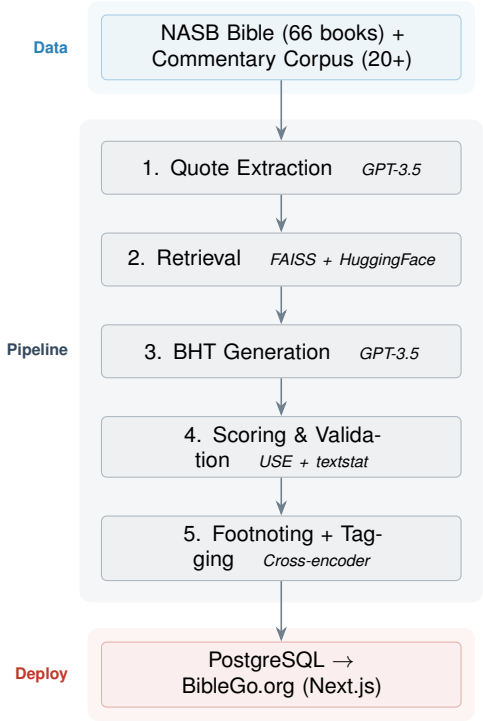

Figure 1: End-to-end system architecture. Commentary and Bible text flow through the five-phase AI pipeline, producing scored BHTs with footnotes that are stored in PostgreSQL and served via BibleGo.org.

Commentary corpus. The commentary collection comprises over 20 curated sources spanning the 16th through 20th centuries. For the New Testament, nine primary commentators form **Tier 1**: Henry Alford, Jamieson-Fausset-Brown, Albert Barnes, Marvin Vincent, John Calvin, Philip Schaff, Archibald T. Robertson, John Gill, and John Wesley. Additional commentators are assigned to Tier 2 and Tier 3 based on their breadth and scholarly standing. For the Old Testament, the corpus includes Keil & Delitzsch, Albert Barnes, Adam Clarke, and specialty commentators such as Arthur W. Pink and Charles H. Mackintosh.

All commentaries were curated and vetted by a group of Christians from the Cleveland, Ohio area, ensuring theological reliability and scholarly quality.

3.2 Commentator Tiering

The tiering system serves a critical role in the generation process. When synthesizing a BHT, the LLM is instructed to draw *primarily* from Tier 1 commentators, *sparingly* from Tier 2, and *minimally* from Tier 3. This weighted approach ensures that the most trusted

and comprehensive commentators dominate the output while allowing supplementary perspectives to enrich the synthesis. Table 1 lists the Tier 1 commentators and their approximate date ranges.

Table 1: Tier 1 New Testament commentators.

Commentator	Era
Henry Alford	1810–1871
Jamieson, Fausset & Brown	1871
Albert Barnes	1798–1870
Marvin R. Vincent	1834–1922
John Calvin	1509–1564
Philip Schaff	1819–1893
Archibald T. Robertson	1863–1934
John Gill	1697–1771
John Wesley	1703–1791

3.3 Prompt Engineering

The system relies on three families of prompts, each iteratively refined through multiple versions:

- **Choicest prompts** (v0.1–v2.2): instruct the LLM to extract 1–3 verbatim quotes from a commentator's text, selecting the most "striking, unique, and enlightening" passages.
- **BHT prompts** (v0.1–v0.8): instruct the LLM to synthesize a constrained paragraph from tiered quotes, enforcing word count, quote proportion, and stylistic rules.
- **Tag commentary prompts** (v0.1–v0.7): instruct the LLM to segment multi-verse commentary into individual verse-level sections.

Prompt versioning enables systematic comparison of prompt effectiveness across generation runs, a form of ablation study applied to prompt design.

3.4 Commentary Digitization

Not all commentaries in the corpus were available in digital form. Two major digitization efforts were undertaken by volunteers to expand the commentary corpus:

Alford's New Testament for English Readers. Henry Alford's four-volume commentary (1868–1872) was digitized from physical book scans totaling 2,544 pages across all 27 New Testament books. The pipeline consisted of three stages: (1) *extraction* via Tesseract OCR and Google Cloud Vision, converting scanned pages to text; (2) *human correction*, in which 28 volunteers spent hundreds of hours reviewing and correcting OCR errors through a coordinated Google Drive spreadsheet; and (3) *automated consolidation*, which detected chapter boundaries, parsed verse-level commentary, performed spell-checking, and produced structured output files. This effort represents a significant scholarly contribution independent of the AI pipeline.

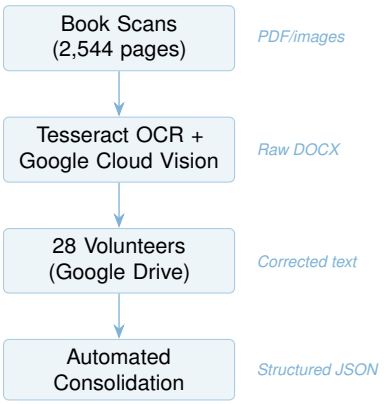

Figure 2: Commentary digitization pipeline used for Alford's and Govett's commentaries.

Govett's Commentary on Revelation. Robert Govett's commentary on Revelation was digitized through the same volunteer-driven process, spanning 627 pages and producing 274 individual verse commentary files covering all 22 chapters. Both digitization projects followed the same three-stage pipeline, demonstrating its replicability for future commentary digitization.

3.5 Original Language Tagging

A separate pipeline was developed to tag every word in the Bible with its corresponding Hebrew or Greek Strong's concordance number. The raw NASB text, with original-language tags provided courtesy of the NASB, was processed through a dedicated parser that produced two outputs:

- **Bible tagged files**: 66 JSON files (one per book), where each English word is linked to its Strong's code (e.g., "beginning" → H7225).
- **Tagged occurrence index**: 14,141 individual Strong's code files containing all 321,621 verse occurrences across the Bible, enabling reverse lookup from any Hebrew or Greek word to every verse where it appears.

A dictionary of 9,333 Hebrew and 5,564 Greek entries was also compiled, providing original-language forms,

transliterations, and definitions. This tagging infrastructure powers the Word Donut feature on BibleGo.org (Section 7) and also enabled the TAG retrieval strategy for OT BHT generation (Section 4).

4 Methods

The BibleGoAI pipeline transforms raw multi-author biblical commentary into concise, source-adherent summary paragraphs called *Brief Helpful Texts* (BHTs). The process unfolds across six sequential phases, each building on the output of the previous. Figure 3 illustrates the generation flow.

4.1 Phase 1: Quote Extraction

Given a target verse, the system loads all available commentary from the curated corpus. Each commentator's text is submitted to GPT-3.5-turbo [OpenAI, 2023] with a *choicest prompt* (iteratively refined from v0.1 to v2.2) that instructs the model to extract one to three "striking, unique, and enlightening" quotes. The prompt enforces strict constraints:

- Quotes must be **verbatim**—no paraphrasing or injected words.
- Each quote must not exceed 100 words.
- Quotes should capture the most distinctive insights, avoiding generic observations.

Extracted quotes are validated against the original commentary to ensure fidelity. Quotes containing words absent from the source material are flagged and re-extracted. This phase produces a `ChoicestQuotesForVerse` object mapping each commentator to their selected excerpts.

4.2 Phase 2: Retrieval

As an alternative or supplement to direct quote extraction, the system supports a retrieval-augmented approach using dense vector search. Commentary texts are split into chunks using one of three strategies:

1. **Character splitting**: Fixed-size chunks with overlap (via LangChain's `CharacterTextSplitter`) [Chase, 2022].
2. **Recursive splitting**: Hierarchical splitting at paragraph, sentence, and word boundaries (`RecursiveCharacterTextSplitter`).
3. **Semantic splitting**: Embedding-aware chunking that respects topical boundaries (`SemanticChunker`).

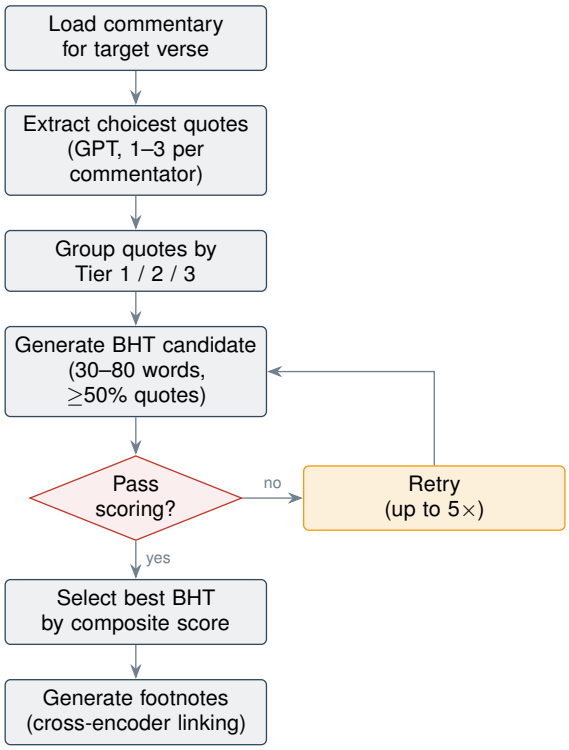

Figure 3: BHT generation flow. Each verse undergoes quote extraction, tiered grouping, constrained generation with up to five retry attempts, quality scoring, and footnote linking.

Chunks are embedded using HuggingFace's `all-mpnet-base-v2` sentence transformer [Song et al., 2020] and indexed with FAISS [Johnson et al., 2019]. At query time, the target Bible verse is embedded and the top-k most semantically similar commentary passages are retrieved, ranked by cosine similarity.

This retrieval mechanism enables coverage of commentaries that span multiple verses, where a single verse may not have a dedicated entry. Four retrieval variants were evaluated: simple character-based RAG, recursive RAG (R-RAG), semantic RAG (S-RAG), and a tagging-based approach (TAG) that uses LLM-driven verse segmentation.

4.3 Phase 3: BHT Generation

The core synthesis step produces a BHT from the extracted or retrieved quotes. Commentators are organized into three tiers (Section 3), and the selected quotes are submitted to GPT-3.5-turbo with a *BHT prompt* (v0.1–v0.8) specifying the following constraints:

- The output must be a single coherent paragraph of

30–70 words in 1–3 sentences.
- At least 50% of the words must originate from the provided quotes.
- The paragraph should draw primarily from Tier 1, sparingly from Tier 2, and minimally from Tier 3.
- No references to commentators by name, verse numbers, or the word "passage."
- The text must not take the form of a list.

The system makes up to five generation attempts per verse, scoring each candidate and retaining the highest-quality result.

4.4 Phase 4: Split-BHT Variant

To explore whether multi-stage generation improves output quality, a *split-BHT* pipeline decomposes generation into two sub-steps:

1. **Split-A**: An initial prompt produces a draft synthesis focused on extracting key themes from the tiered quotes.
2. **Split-B**: A refinement prompt polishes the Split-A output into a final BHT, enforcing word count, quote proportion, and stylistic constraints.

Both sub-prompts were independently versioned and evaluated. This approach trades additional API calls for potentially higher coherence and fidelity.

4.5 Phase 5: Scoring and Validation

Every generated BHT undergoes a multi-dimensional quality assessment combining hard constraints with continuous metrics.

Hard constraints.
- Word count within 30–80 (strict range: 20–100).
- Quote token proportion between 30% and 90%.
- Absence of forbidden tokens (commentator names, verse references, "passage").
- No list formatting.

Continuous metrics.
- **Commentary accuracy**: Tier-weighted cosine similarity between the BHT and source quotes, computed via the Universal Sentence Encoder [Cer et al., 2018]. A minimum score of 2.3 is required.
- **Verse accuracy**: Cosine similarity between the BHT and the target verse embedding, measuring topical relevance.
- **Readability**: Flesch-Kincaid Grade Level, Flesch Reading Ease, Gunning Fog Index, Automated Readability Index, and Dale-Chall score [Kincaid et al., 1975, Gunning, 1952].
- **Tier distribution**: The proportion of BHT content attributable to each commentator tier, measured via per-tier semantic similarity.

Among the up to five generation attempts, the BHT with the highest composite quality score is selected. The scoring function prioritizes commentary accuracy while penalizing outputs that violate hard constraints.

4.6 Phase 6: Footnoting and Tagging

The final phase establishes provenance links between the generated BHT and its source commentaries.

Footnoting. Each sentence in the BHT is compared against all source quotes using cross-encoder scoring via `sentence-transformers` [Reimers and Gurevych, 2019]. For each sentence, the system identifies the most similar quote and generates a footnote containing the commentator name, quote index, similarity score, and source location. This enables readers to trace any claim back to its originating commentary.

Commentary tagging. For commentaries spanning multiple verses, a *tag commentary prompt* (v0.1–v0.7) instructs GPT to segment the text into per-verse sections. The output is a JSON structure mapping each verse number to its relevant content, enabling verse-level retrieval from range-based sources.

4.7 The OT Challenge

The New Testament pipeline described above was designed for commentary that is organized verse-by-verse, which is the norm for NT commentators. After the public release of NT BHTs, the team turned to the Old Testament—and encountered a fundamental structural difference: OT commentary is rarely organized by individual verse. Instead, commentators typically discuss passages spanning multiple verses, making direct quote extraction for a single verse impractical.

Two initial approaches were explored:

1. **Commentary "versification"**: Using GPT to tag multi-verse commentary with per-verse labels, effectively segmenting it into verse-level sections before

applying the standard NT pipeline. While conceptually appealing, the accuracy of automated verse tagging proved inconsistent at scale.

2. **Enhanced choicest extraction**: Modifying the choicest prompt to extract verse-specific quotes directly from multi-verse commentary, bypassing the need for segmentation. This also proved unreliable, as the model struggled to isolate verse-relevant content from longer passages.

Both approaches were evaluated over several weeks. The breakthrough came when an AI/ML engineer suggested using retrieval-augmented generation: rather than requiring verse-level segmentation, the system could embed commentary chunks and retrieve the most semantically relevant passages for each verse via vector search. This RAG-based approach (Phase 2) produced satisfactory results and unlocked OT BHT generation at scale, ultimately enabling completion of the full Bible corpus.

5 Evaluation

We evaluate BibleGoAI across multiple dimensions: aggregate quality metrics, generation strategy comparison, generation attempt analysis, and readability. All metrics are computed from the full corpus of generated BHTs: 7,957 New Testament verses and 10,492 Old Testament verses (regular variant), with additional comparison across Split-A (12,653 verses) and Split-B (12,653 verses) variants.

5.1 Aggregate Quality Metrics

Table 2 summarizes the key metrics across the NT and OT corpora. Quality scores reflect the composite tier-weighted commentary accuracy, while quote token proportion measures the fraction of BHT words drawn directly from source commentary.

Table 2: Aggregate BHT quality metrics across testaments. Values are mean $\pm$ standard deviation.

Metric	**NT** (n=7,957)	**OT** (n=10,492)
Quality Score	1.912 ± 0.348	1.785 ± 0.282
Word Count	84.2 ± 11.1	62.9 ± 11.6
Quote Proportion (%)	39.6 ± 7.4	31.8 ± 6.3
Verse Accuracy	—	0.417 ± 0.147

The NT corpus achieves higher average quality scores (1.912 vs. 1.785), likely reflecting the more extensive commentary coverage for New Testament verses and the maturity of the NT prompt versions. NT BHTs are also longer on average (84 vs. 63 words) with higher quote proportions (39.6% vs. 31.8%), suggesting richer source material for synthesis. Verse accuracy was computed only for the OT corpus; see below for discussion.

Verse accuracy (OT only). The *verse accuracy* metric—cosine similarity between the BHT and the target-verse embedding—was computed only for the OT corpus in the reported results. Because the NT and OT pipelines differ (direct per-verse extraction for NT vs. retrieval-augmented generation for OT; see Section 4.7), the OT verse-accuracy value (0.417 ± 0.147) should be read as a sanity check on the OT RAG pipeline rather than as directly comparable to a would-be NT value. Computing this metric retrospectively across the full NT corpus is straightforward and is noted as near-term follow-up work.

5.2 Generation Strategy Comparison

We compare four retrieval and generation strategies applied to OT verses, where the same underlying commentary corpus is processed through different pipelines:

- **Regular**: Standard single-pass generation from tagged commentary.
- **Split-A**: First stage of the two-stage split pipeline (draft synthesis).
- **Split-B**: Second stage refinement of Split-A output.
- **RAG variants**: Character-based, recursive, and semantic retrieval.

Figure 4 compares the average quality scores across strategies.

Table 3: Generation strategy comparison across OT verses.

Metric	**Regular**	**Split-A**	**Split-B**
n (verses)	10,492	12,653	12,653
Quality Score	1.785	**1.935**	1.710
Verse Accuracy	0.417	**0.467**	0.353
Word Count	62.9	55.1	30.8
Quote Prop. (%)	31.8	32.4	32.6

Split-A achieves the highest quality scores (1.935) and verse accuracy (0.467), outperforming both the regular single-pass approach and the Split-B refinement stage. Notably, Split-B produces significantly shorter BHTs (30.8 words on average) with lower scores, suggesting

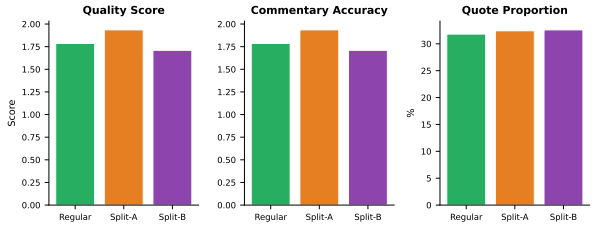

Figure 4: Average quality scores by generation strategy across OT verses.

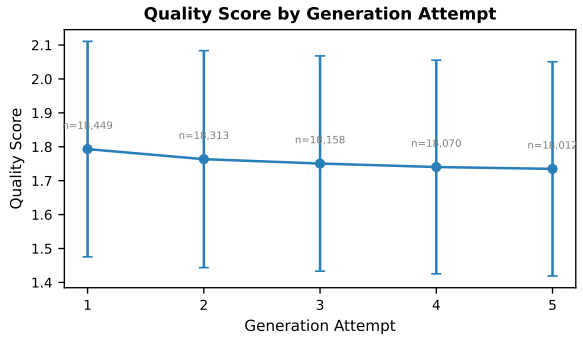

Figure 5: Average quality score by generation attempt number (1–5). Later attempts show diminishing returns.

that the refinement step's aggressive compression may sacrifice content quality. The regular strategy offers a balance between length and quality.

5.3 Generation Attempt Analysis

The system generates up to five candidate BHTs per verse and selects the highest-scoring attempt. Figure 5 shows the average quality score by attempt number, revealing whether iterative generation yields meaningful improvement.

5.4 Commentator Tier Contribution

Figure 6 shows the distribution of semantic similarity across the three commentator tiers, illustrating how well the tiering instructions are followed during generation.

5.5 Quality Score Distribution

Figure 7 presents the distribution of quality scores across all NT and OT verses, showing the overall consistency and range of the system's output.

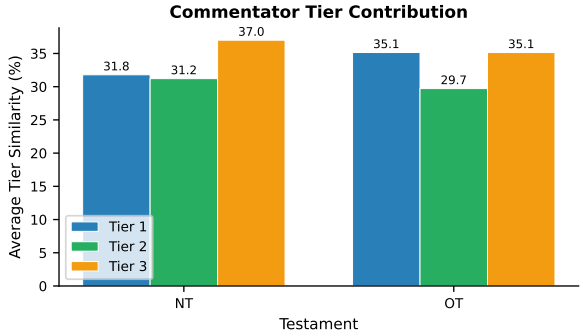

Figure 6: Average commentator tier similarity scores. Tier 1 consistently contributes the most to BHT content.

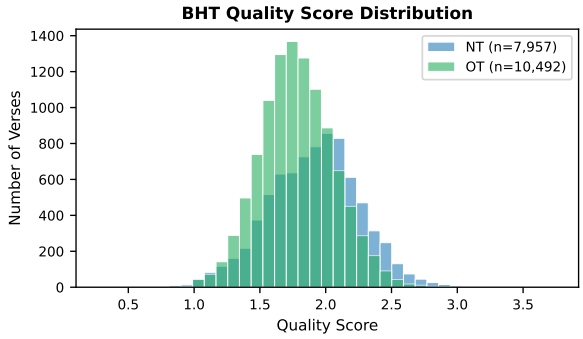

Figure 7: Distribution of BHT quality scores across NT and OT corpora.

5.6 Human Evaluation

In addition to automated metrics, BHT quality was evaluated through six rounds of structured user testing conducted with members of the project's Bible study community:

- **BHT V2 Quality Test**: Human raters scored NT BHTs on a normalized quality scale. Scores ranged from 50.4 to 94.2 out of 100, with a mean of 67.5. The highest-rated BHT (Romans 6:7, score 94.2) demonstrated strong synthesis of commentary insights, while lower-scoring BHTs guided subsequent prompt refinements.
- **OT BHT V1 Quality Test**: A separate round evaluated the first generation of OT BHTs, providing feedback that informed the comparison of RAG versus tagging strategies.

Table 4 presents the human quality scores from the BHT V2 evaluation.

These human scores, combined with the automated quality metrics, provided a feedback loop that drove iterative prompt refinement—most notably the May 2024 prompt

Table 4: Human evaluation scores (BHT V2 Quality Test).

Verse	Quality Score
Romans 6:7	94.2
John 5:22	80.1
Ephesians 1:22	77.3
Ephesians 6:13	65.5
Galatians 5:22	62.7
John 10:27	57.0
Romans 6:21	54.1
Revelation 3:3	50.4
Mean	**67.7**

updates that shifted the choicest extraction focus from "striking and enlightening" to "helpful, explanatory, and illuminating" quotes targeting basic words and facts in each verse.

6 Discussion

Strengths. The multi-stage pipeline design offers several advantages over end-to-end generation. By extracting verbatim quotes before synthesis, the system creates an auditable intermediate representation that can be inspected and validated independently of the final BHT. The tiered commentator system ensures that the most authoritative voices dominate the output, while the constraint enforcement (word count, quote proportion, forbidden tokens) prevents many common failure modes of unconstrained LLM generation.

The iterative prompt refinement process—spanning over 20 prompt versions across three prompt families—proved essential. Early prompt versions (v0.1–v0.3) produced BHTs that frequently exceeded word limits, included commentator names, or adopted list formatting. By v0.8 (BHT) and v2.2 (choicest), these issues were largely eliminated, demonstrating the value of systematic prompt engineering with quantitative feedback loops.

Limitations. Several limitations warrant discussion. First, the entire deployed corpus was generated using GPT-3.5-turbo, which was among the most capable generally available models at the time of development. A small, informal comparison on 13 verses against Llama 3.3-70B (via Groq) and GPT-4o-mini suggested broadly comparable quality, but the sample is too small to support general claims about model impact, and a full-scale regeneration and re-evaluation on newer models remains future work. Second, the quality scoring framework, while multi-dimensional, relies heavily on embedding-based similarity, which may not capture all aspects of theological accuracy or interpretive source adherence. Third, the 50% quote proportion constraint, while ensuring grounding, can sometimes produce BHTs that read as stitched-together phrases rather than naturally flowing prose. Fourth, we do not yet include a zero-shot LLM baseline—e.g., prompting a frontier model directly with all available commentary for a verse and asking it to summarize—so the marginal value of the multi-stage pipeline over modern end-to-end generation has not been quantitatively isolated.

Source adherence vs. fluency. A fundamental tension exists between source adherence and fluency of the generated text. Higher quote proportions ensure adherence but can reduce coherence; lower proportions allow smoother prose but risk introducing LLM-generated content that may not reflect the commentators' actual positions. Our constraint of 50% represents a pragmatic balance, validated through iterative testing and user feedback.

Prompt engineering insights. The evolution of prompts yielded several practical insights: (1) explicit negative constraints ("do not mention commentators") were more effective than positive-only instructions; (2) specifying exact word count ranges reduced variance more than vague length instructions; (3) the tiering language ("primarily," "sparingly," "minimally") meaningfully influenced the tier distribution in generated outputs; and (4) requesting numbered quotes in the extraction phase significantly improved downstream validation.

Project process. Several practical lessons emerged from the collaborative development process. The team's request to use the English Standard Version (ESV) was denied by the publisher, leading to the selection of the NASB 1995 with permission from the Lockman Foundation—a decision that proved beneficial given the NASB's reputation for literal translation. Over 18 demo presentations were prepared during an intensive April–May 2024 development period, each showcasing incremental improvements in generation quality, retrieval strategies, and UI features. These presentations,

preserved in the project's demo archive, provided a structured mechanism for gathering feedback and tracking progress through the weekly round table meetings.

7 The BibleGo.org Platform

The BHTs generated by BibleGoAI are served to end users through **BibleGo.org**, a web application built with Next.js [Vercel, 2023], React, and TypeScript, backed by a PostgreSQL database and hosted on Vercel. The platform currently serves hundreds of users per month and provides three primary modes of interaction.

Commentary reading. The main reading interface presents a responsive three-column layout (Figure 8): a book selector on the left, the Bible text (NASB) in the center with inline original-language tags, and a commentary panel on the right. When a user selects a verse, the corresponding BHT is displayed along with footnotes that link each sentence to its source commentary. Users can expand individual footnotes to view the original quotes, the commentator attribution, and a similarity score indicating the strength of the connection. Each commentator's full excerpt is also available as a separate card, enabling side-by-side comparison of perspectives. On mobile devices, the commentary panel appears as a slide-up drawer (Figure 9).

Word study. BibleGo.org includes an interactive word study feature—the *Word Donut*—powered by Strong's concordance data built from the original language tagging pipeline (Section 3). Users can select any word in the Bible text to view its original Greek or Hebrew form, transliteration, and definition. An interactive doughnut chart then visualizes how the NASB translators chose to render that word across all its occurrences in the Bible, showing each English translation variant and its frequency (Figure 10). This feature supports pagination and filtering by translation variant, enabling original-language exploration without requiring knowledge of Greek or Hebrew.

Full commentary browsing. Users can switch from the BHT summary view to a full commentary mode, which displays the complete chapter-level commentary for any individual commentator (Figure 12). This enables readers who encounter an interesting insight in a

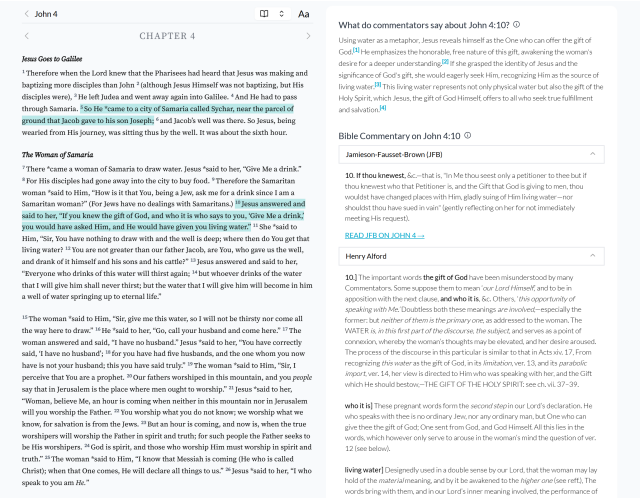

Figure 8: The BibleGo.org desktop reading interface. The three-column layout presents book navigation (left), the NASB Bible text with inline original-language tags (center), and the BHT commentary panel (right) with footnotes linking each sentence to its source commentator.

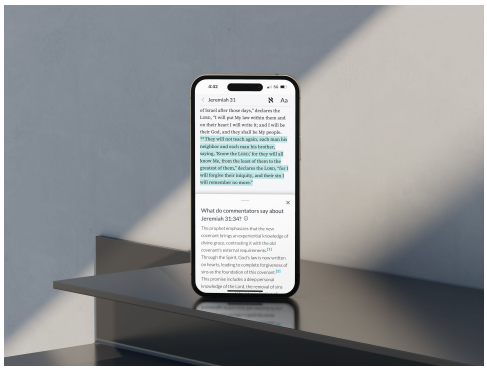

Figure 9: The BibleGo.org mobile experience with the BHT commentary drawer. Footnotes (numbered markers) link each sentence back to its source commentator, allowing users to trace every claim to the original commentary.

BHT to dive deeper into the full context of that commentator's work.

User experience. The application features dark mode support, adjustable font sizes (persisted across sessions), mobile-responsive design with a drawer-based layout for smaller screens, deep linking via URL parameters (e.g., `/read?v=Matthew.1.1`), and an in-app feedback system with structured categories. The UI underwent four rounds of dedicated usability testing—including

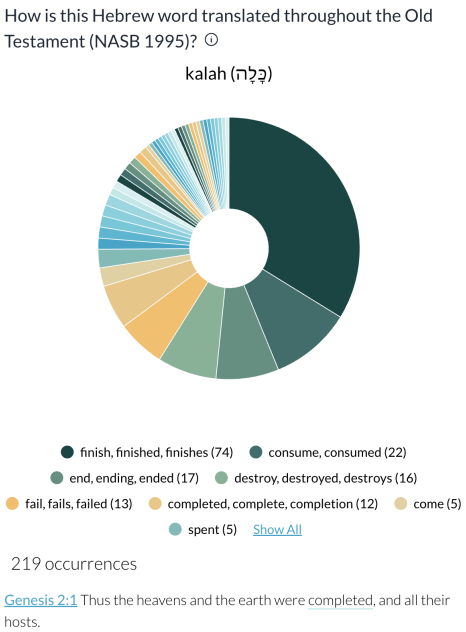

Figure 10: The Word Donut visualization for the Hebrew word *kalah*, showing how it is translated across 219 Old Testament occurrences. Segments represent distinct English renderings (e.g., "finish" 74×, "consume" 22×), enabling users to explore semantic range without knowledge of the original language.

Figure 11: Mobile view of the Word Donut for the Hebrew word *racham* ("have compassion") in Isaiah 55. The doughnut chart shows translation frequencies across the Old Testament, enabling word study directly from the reading interface.

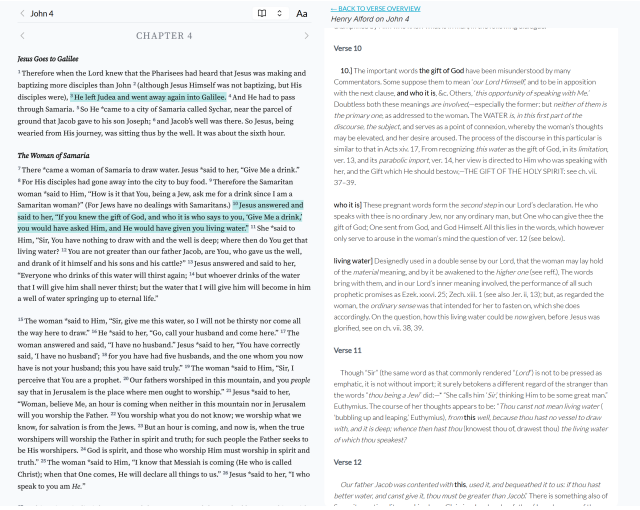

Figure 12: The full commentary browsing mode, displaying a complete chapter-level commentary from an individual commentator. Users can explore the full original text that informs each BHT.

tests focused specifically on the drawer navigation and the word study tool—with feedback directly informing design iterations.

Data flow. Pre-generated BHTs from the BibleGoAI pipeline are loaded into a PostgreSQL database. The Next.js application queries this database through server-side API routes, returning BHT text, footnotes, scoring metadata, and commentator information. Bible text with Strong's number tags is served from static JSON files, while Strong's concordance entries are queried from the database. This hybrid approach balances query flexibility with serving performance.

8 Conclusion and Future Work

We have presented BibleGoAI, a multi-stage system for synthesizing concise, source-adherent summaries of biblical commentary using large language models, retrieval-augmented generation, and semantic similarity scoring. The system transforms a curated corpus of over 20 commentators spanning five centuries into 30–80 word Brief Helpful Texts (BHTs) that are grounded in source material, quality-scored across multiple dimensions, and traceable via automated footnotes.

Our evaluation across 7,957 NT and 10,492 OT verses demonstrates that the pipeline produces consistent, high-quality summaries with strong commentary accuracy

and appropriate tier distribution. The comparison of four generation strategies and three LLM backends provides practical guidance for similar systems. The deployed application at BibleGo.org, serving hundreds of monthly users, validates the real-world utility of the approach.

Future work. Several directions remain for future development:

- **Complete OT coverage**: Extending BHT generation to all 39 Old Testament books (currently 24 are covered).
- **Additional commentators**: Incorporating more modern and diverse theological perspectives into the corpus.
- **User studies**: Conducting formal user studies to evaluate BHT quality, usefulness, and theological accuracy as perceived by Bible study practitioners.
- **Enhanced original-language tools**: Expanding the word study features to include Hebrew morphology, cross-references, and parallel passage analysis.
- **Multilingual support**: Generating BHTs in languages beyond English to serve the global Bible study community.
- **Fine-tuned models**: Training domain-specific models on biblical commentary to reduce dependency on general-purpose LLMs and potentially improve both fidelity and cost efficiency.

By demonstrating that LLMs can be effectively constrained, scored, and validated to produce source-adherent scholarly summaries, BibleGoAI contributes both a practical tool for Bible study and a methodological framework applicable to other domains where multi-source synthesis with provenance tracking is required.

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References

Daniel Cer, Yinfei Yang, Sheng-yi Kong, Nan Hua, Nicole Limtiaco, Rhomni St. John, Noah Constant, Mario Guajardo-Cespedes, Steve Yuan, Chris Tar, et al. Universal sentence encoder. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 169–174, 2018.

Harrison Chase. Langchain. https://github.com/langchain-ai/langchain, 2022.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

Robert Gunning. *The Technique of Clear Writing*. McGraw-Hill, 1952.

Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2019.

J. Peter Kincaid, Robert P. Fishburne, Richard L. Rogers, and Brad S. Chissom. Derivation of new readability formulas for navy enlisted personnel. *Research Branch Report 8-75, Naval Technical Training Command*, 1975.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

NLLB Team. No language left behind: Scaling human-centered machine translation. In *arXiv preprint arXiv:2207.04672*, 2022.

OpenAI. GPT-3.5 turbo. https://platform.openai.com/docs/models/gpt-3-5-turbo, 2023.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992, 2019.

Kaitao Song, Xu Tan, Tao Qin, Jianfeng Lu, and Tie-Yan Liu. MPNet: Masked and permuted pre-training for language understanding. In *Advances in Neural Information Processing Systems*, volume 33, pages 16857–16867, 2020.

Ashwin Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 2017.

Vercel. Next.js: The react framework for the web. https://nextjs.org/, 2023.